

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9.	(a)			mass of NO ₂ = 25.0 × 0.714 ÷ 100 = 0.1785 (1) moles NO ₂ = 0.1785 ÷ 46 = 3.88 × 10 ⁻³ (1) answer must be to 3 sig figs to gain full marks		2		2	2	
	(b)	(i)		$K_c = \frac{[NO_2]^2}{[N_2O_4]}$ (1) unit mol dm ⁻³ (1) allow ecf for unit		2		2	1	
		(ii)	I	concentration of NO ₂ = 2 × (0.400 – 0.0581) = 0.6838 (1) value of K _c = 8.05 (1) solvent is CCl ₄ (1) (allow ecf)		2	1	3	1	
			II	CS ₂ given with any valid reason (1) explanation in terms of position of equilibrium or amount of product formed (1)		1	1	2		

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					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		catalyst allows a lower temperature to be used (1) this leads to an increased yield (1) (must get first mark)		1	1	2		
		(ii)		a higher pressure should be used (1) shifts equilibrium to the right as this has fewer gas molecules (1) a lower temperature should be used (1) shifts equilibrium to the right as this is the exothermic direction (1) (allow one explanation mark if reference to equilibrium shifting to oppose change with no further detail) accept alternative remove product as it forms (1) equilibrium moves further towards the right / product (1)		1 1	1 1	4		
				Question 9 total	0	10	5	15	4	0